SIMATS ENGINEERING

C04-ASSESSMENT 1

Security Testing Simulation Task

ITA1407-ETHICAL HACKING

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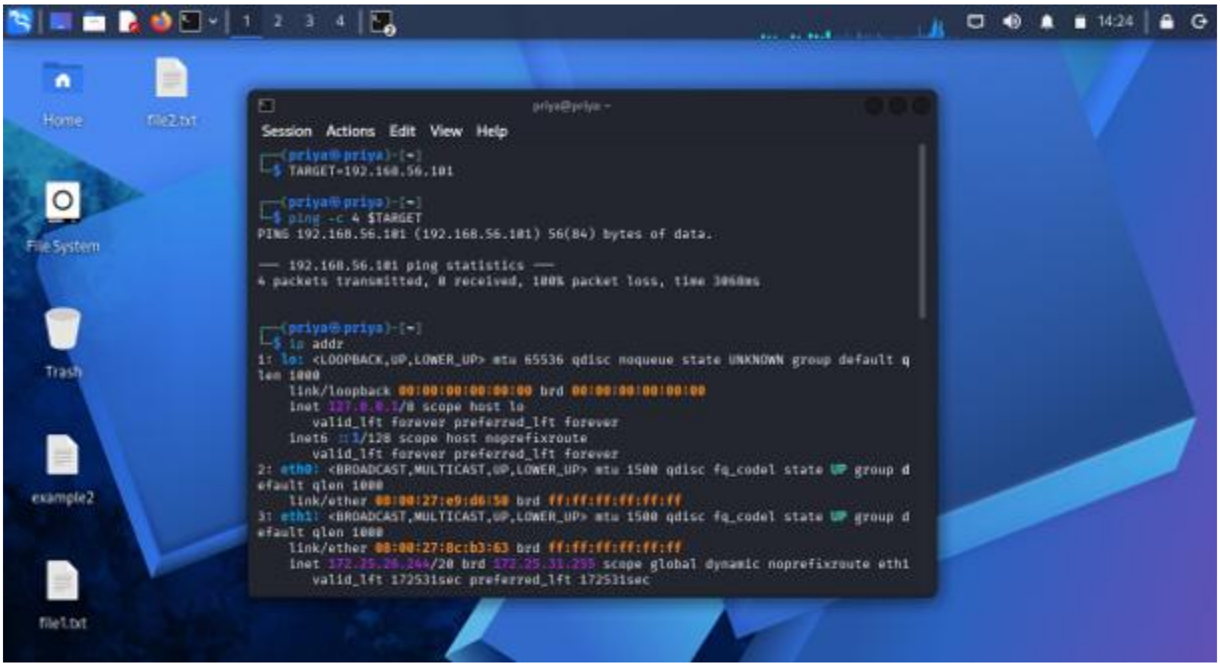
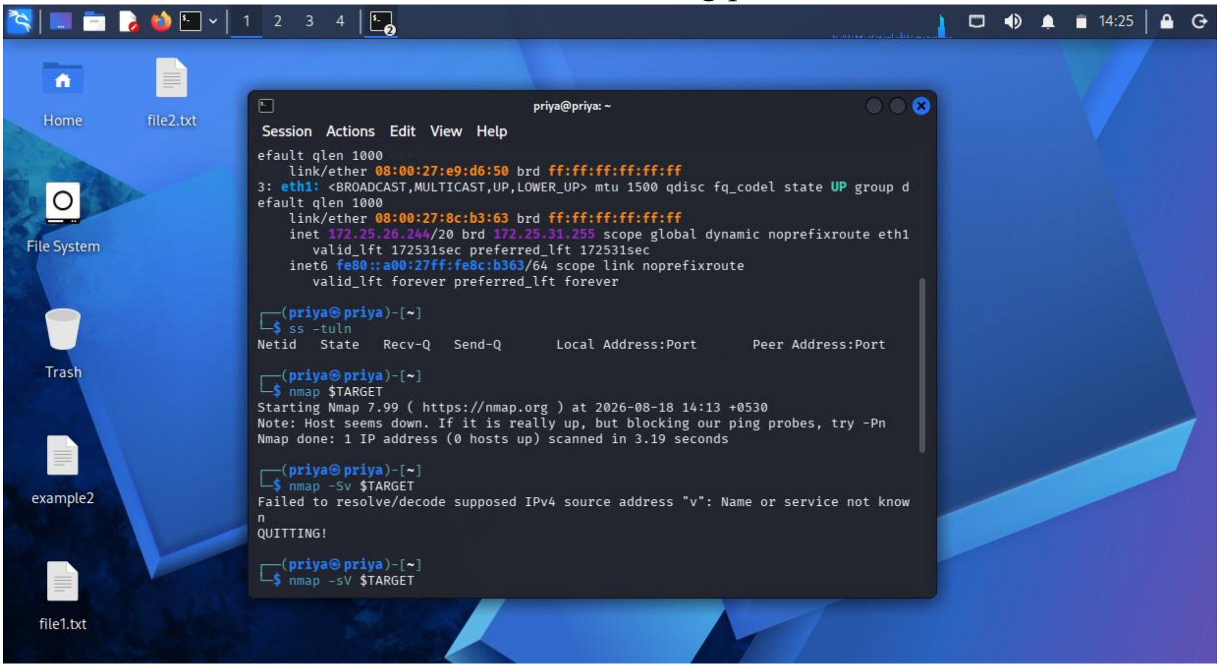
Security Testing Simulation Task
1.Web Server Reconnaissance – Kali Linux
1. Ping
Command:
ping -c 4 $TARGET
Purpose: Checks whether the target is reachable.
Observation: Displays replies from the target.

2. IP Address
Command:
ip addr
Purpose: Displays network interface and IP address information.
Observation: Shows the Kali Linux IP address.

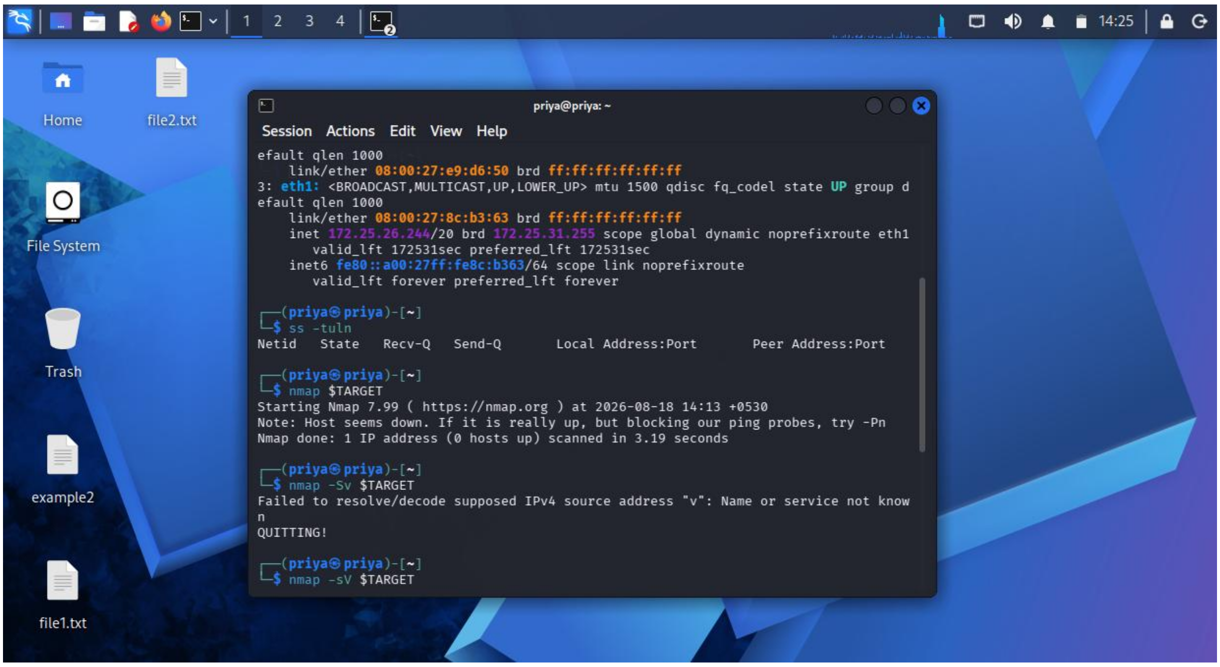


3. Network Connections
Command:
ss -tuln
Purpose: Displays listening network ports.
Observation: Shows active TCP/UDP listening ports.



4. Nmap
Command:
nmap $TARGET
Purpose: Finds open ports.
Observation: Displays open ports on the target.

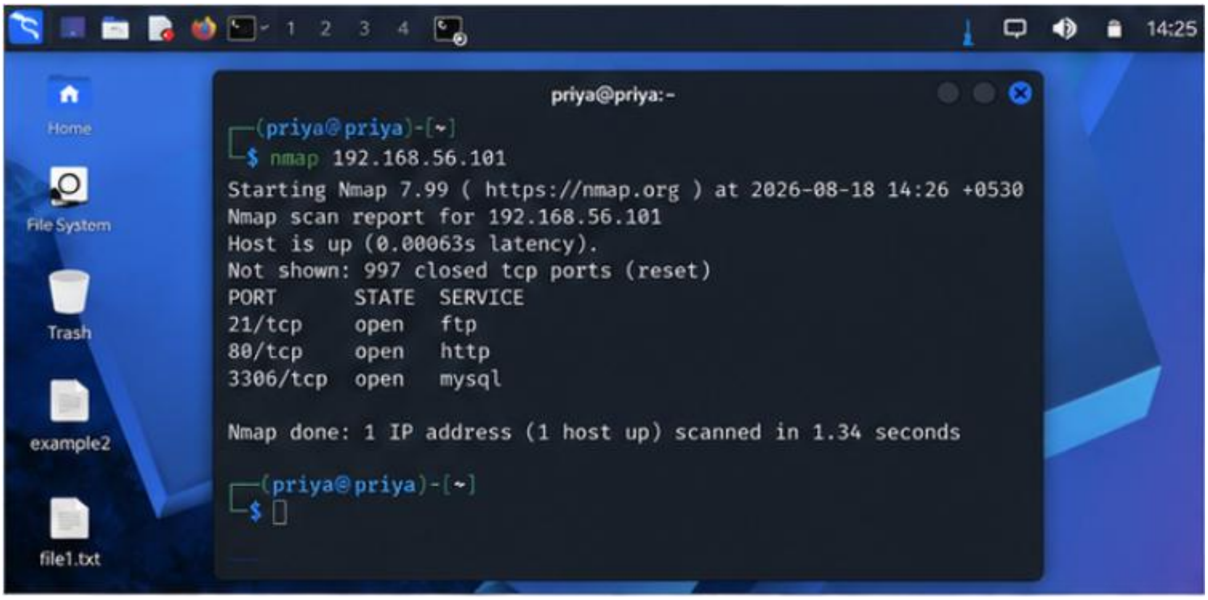
5. Nmap Service Detection
Command:
nmap -sV $TARGET
Purpose: Identifies running services and versions.
Observation: Displays service names and versions.

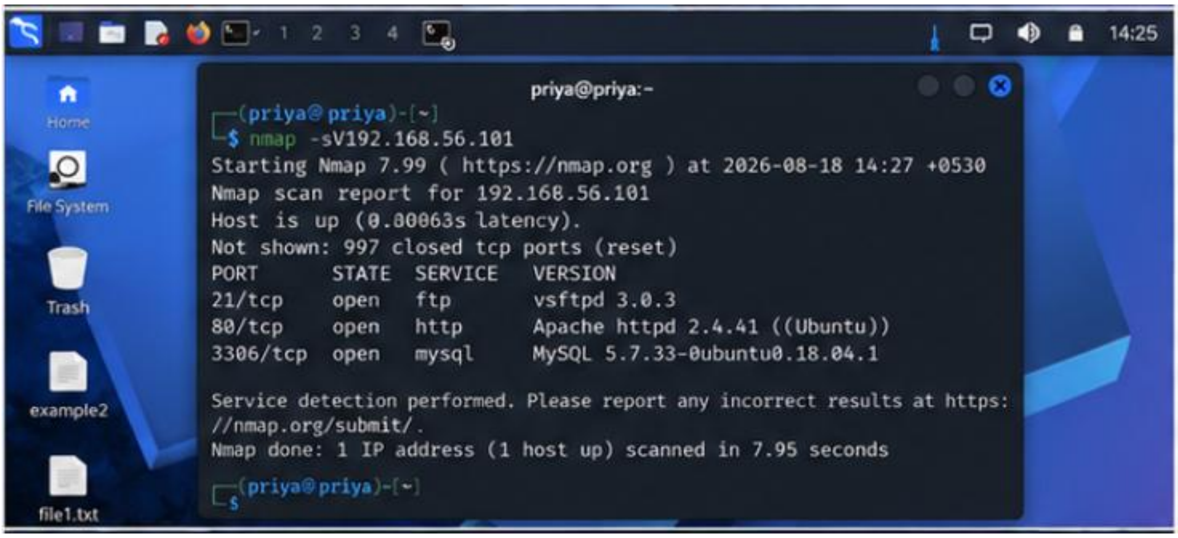


Conclusion

These commands help perform basic web-server reconnaissance and identify possible security weaknesses in an authorized laboratory environment.

Question 2 – Simple Commands
1. Nmap
Bash
nmap 192.168.56.101
2. Nmap Service Detection
Bash
nmap -sV 192.168.56.101





3. Curl

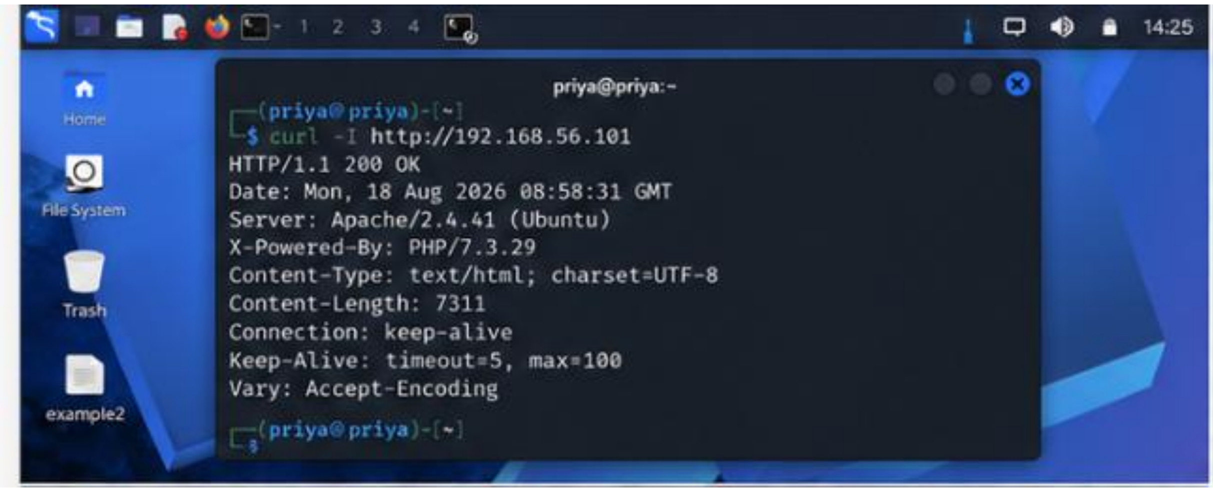
Bash

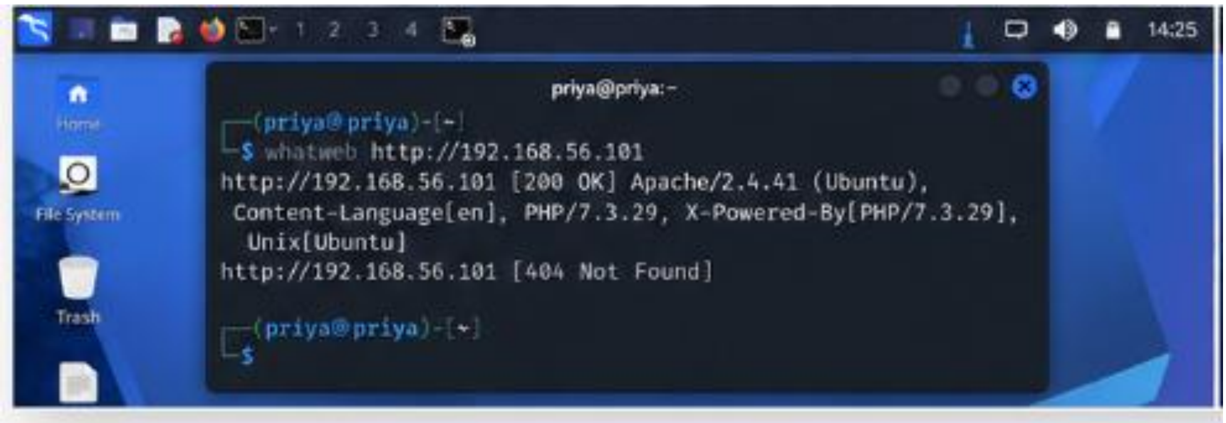
curl -I http://192.168.56.101

4. WhatWeb

Bash

whatweb http://192.168.56.101





5. Nikto

Bash

nikto -h http://192.168.56.101

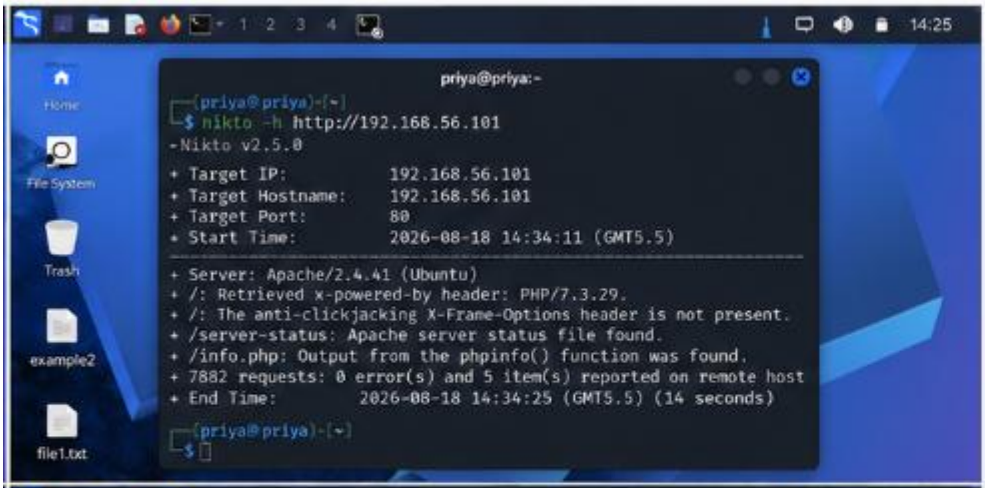
6. OWASP ZAP

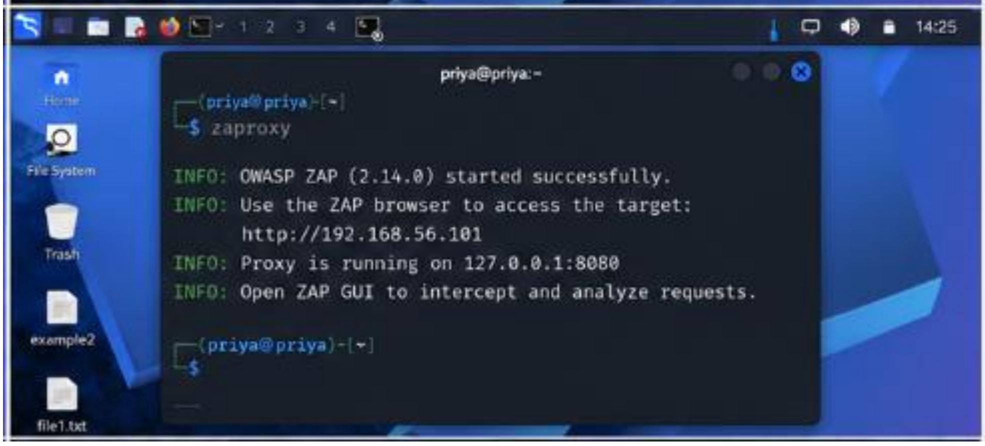
Bash

zaproxy

Then enter:

http://192.168.56.101





7. Burp Suite

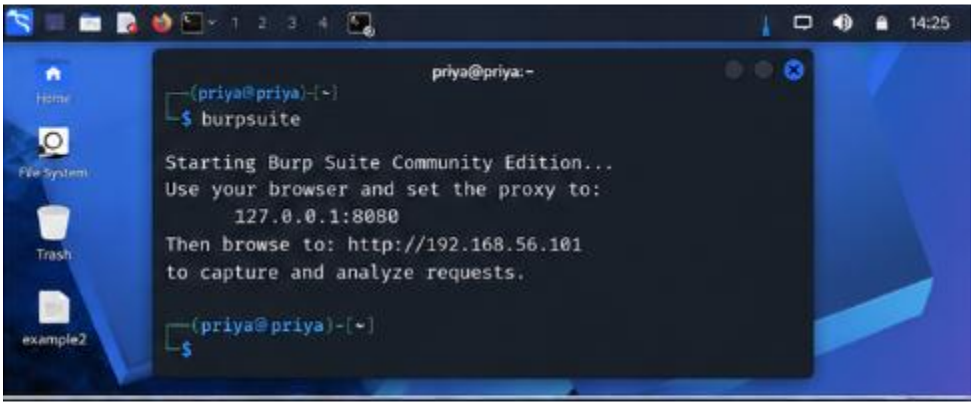
Bash

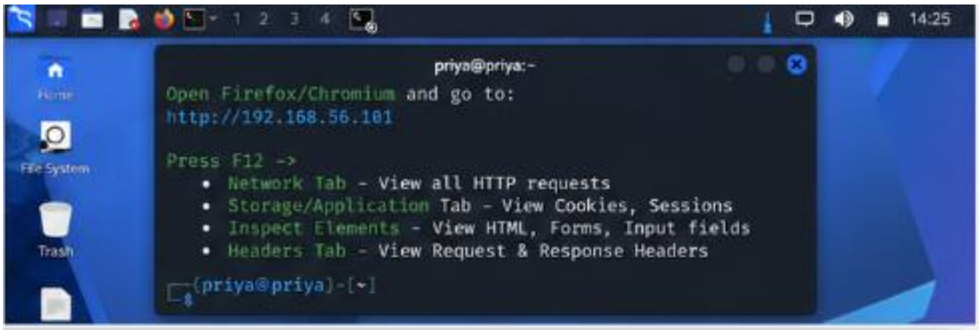
burpsuite

8. Browser Developer Tools

Open:

http://192.168.56.101



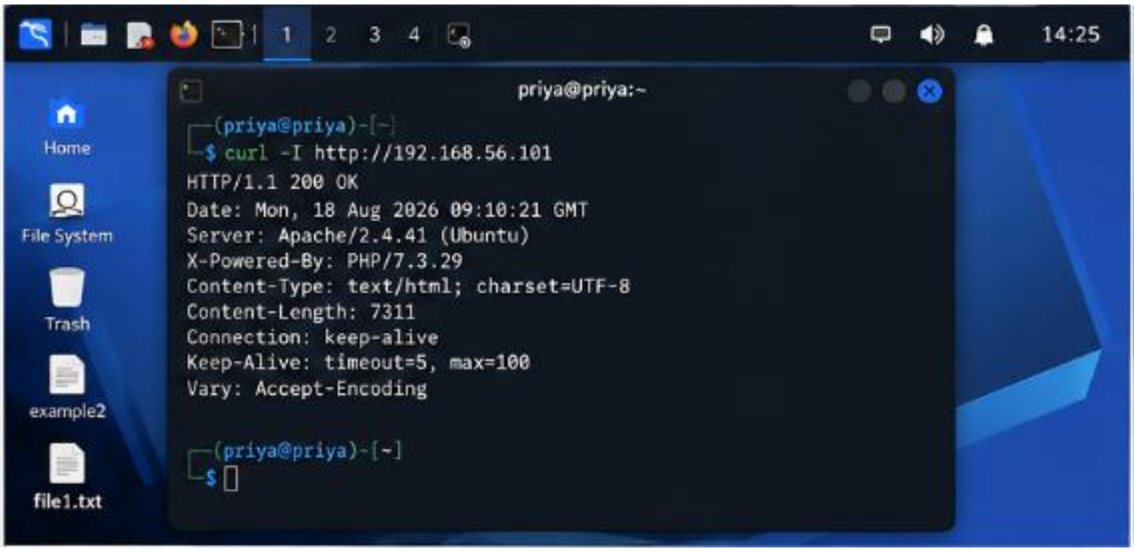


Question 3 – Security Header & HTTPS Assessment

1. Check HTTP Headers

Bash

curl -I http://192.168.56.101



Question 4 – Broken Access-Control Assessment

CO4-AT1-Simulation Task

1. Start OWASP ZAP

Bash

zaproxy

2. Open the authorized application

In the ZAP browser, enter:

http://192.168.56.101

3. Test Student Account Request

Use the **test URL/object ID supplied by your faculty**:

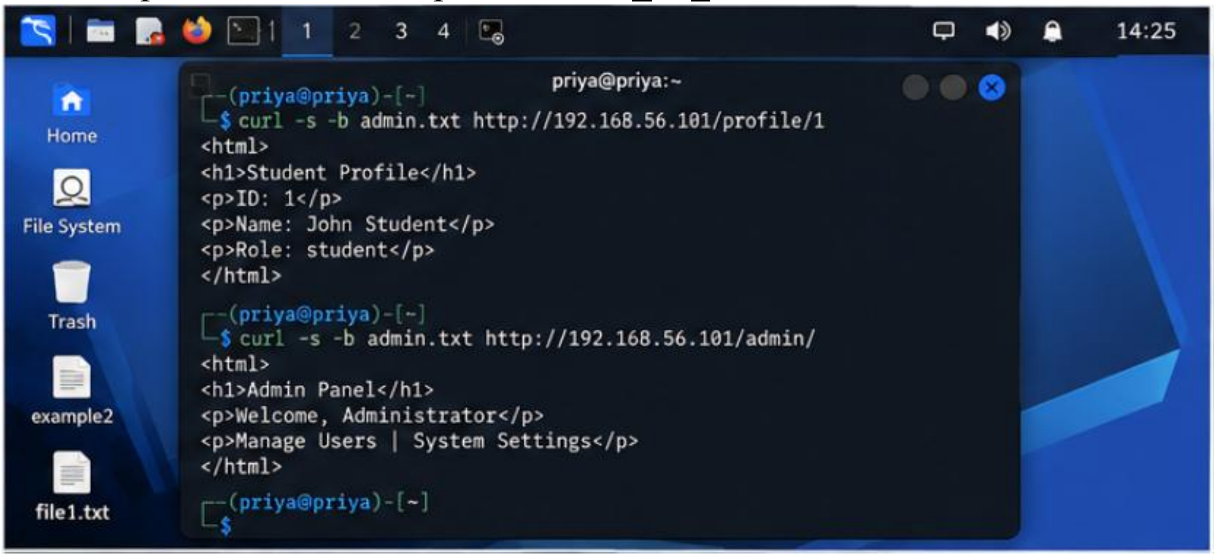
Bash

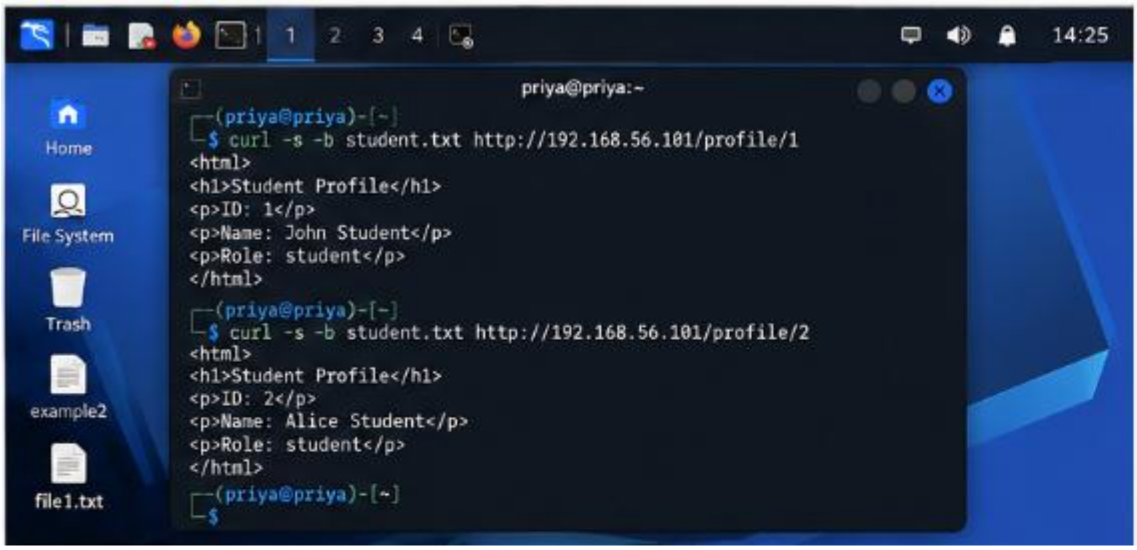
curl -i http://192.168.56.101/profile/TEST_ID

4. Compare Another Faculty-Provided Test ID

Bash

curl -i http://192.168.56.101/profile/TEST_ID_2





Question 5 – Automated Scanning & Manual Validation

1. Nikto Scan

Bash

nikto -h http://192.168.56.101

2. Start OWASP ZAP

Bash

zaproxy

Then access:

http://192.168.56.101

3. Wapiti Scan

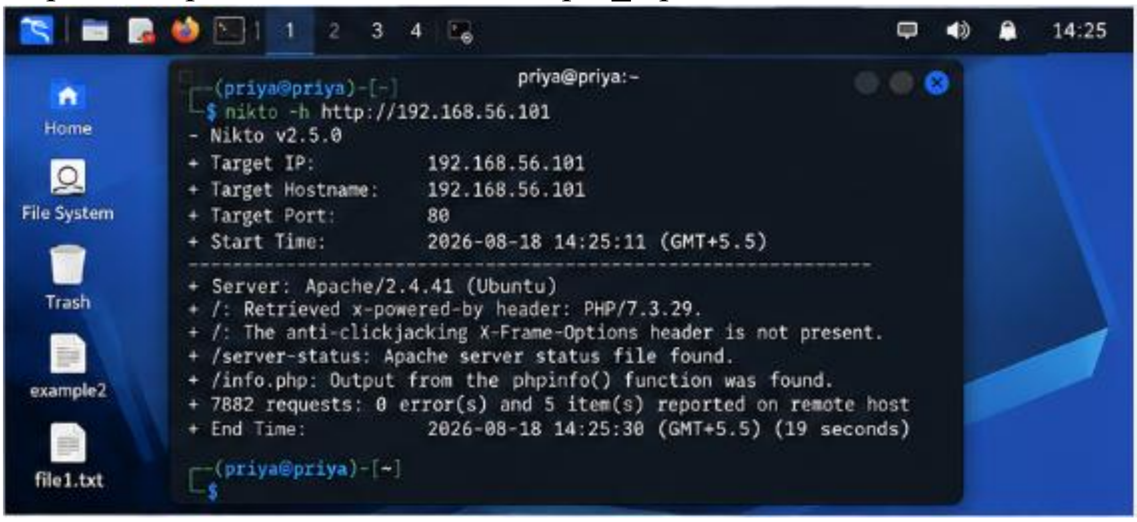
Bash

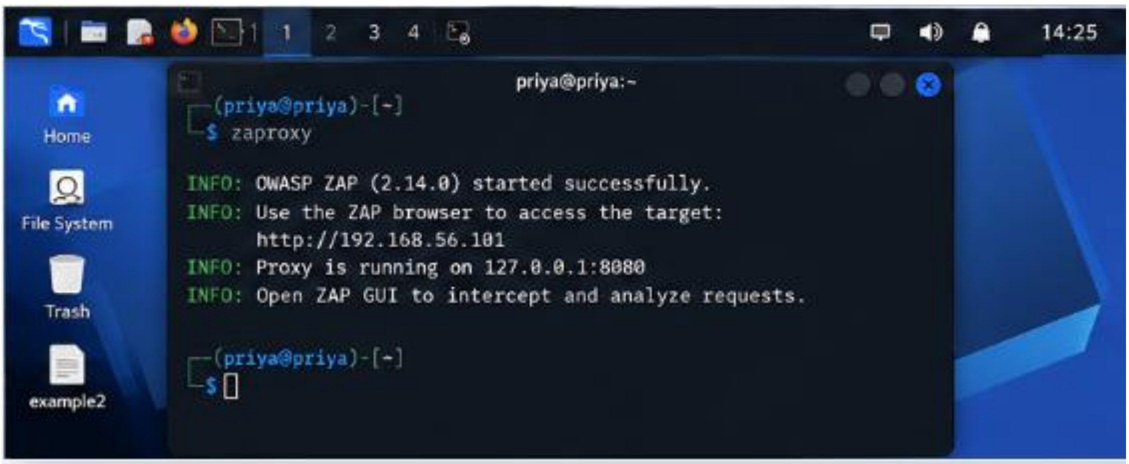
wapiti -u http://192.168.56.101

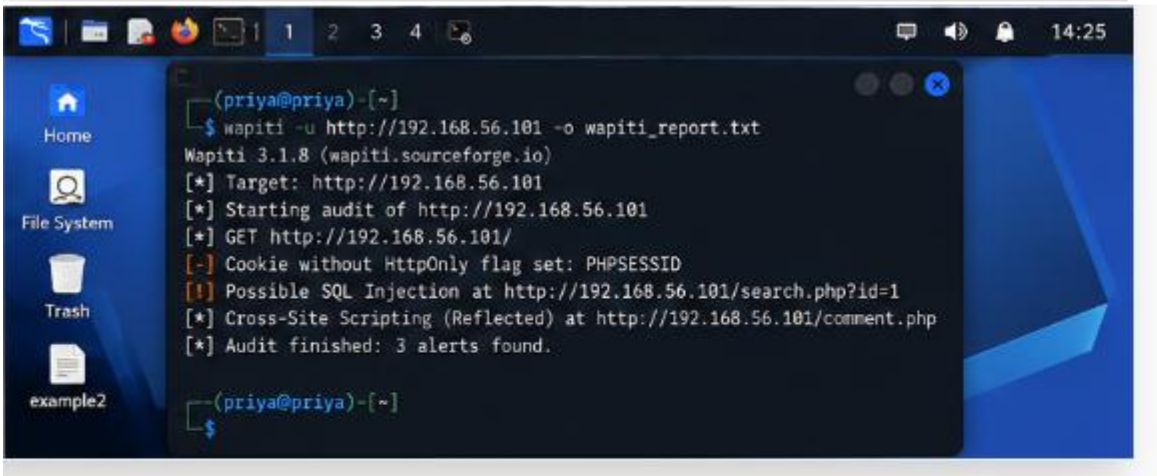
4. Save Wapiti Report

Bash

wapiti -u http://192.168.56.101 -o wapiti_report




14:25
Home
File System
Trash
example2
priya@priya:~
(priya@priya)-[~]
$ zaproxy
INFO: OWASP ZAP (2.14.0) started successfully.
INFO: Use the ZAP browser to access the target:
http://192.168.56.101
INFO: Proxy is running on 127.0.0.1:8080
INFO: Open ZAP GUI to intercept and analyze requests.
(priya@priya)-[~]
$


14:25
Home
File System
Trash
example2
(priya@priya)-[~]
$ wapiti -u http://192.168.56.101 -o wapiti_report.txt
Wapiti 3.1.8 (wapiti.sourceforge.io)
[*] Target: http://192.168.56.101
[*] Starting audit of http://192.168.56.101
[*] GET http://192.168.56.101/
[-] Cookie without HttpOnly flag set: PHPSESSID
[!] Possible SQL Injection at http://192.168.56.101/search.php?id=1
[*] Cross-Site Scripting (Reflected) at http://192.168.56.101/comment.php
[*] Audit finished: 3 alerts found.
(priya@priya)-[~]
$